

Project Overview and ML Approach

1. About project

This project is a **Smart Text Response Classifier** built using Python, Flask, and Machine Learning.

The application accepts a user's text and performs two classification tasks:

- **Language Detection:** Identifies whether the text is English or Hinglish.
- **Category Detection:** Classifies the text into one of 10 categories: Health, Mood/Emotion, Work/Office, Travel, Education, Food, Technology, General/Other, Shopping, and Finance.

The application also returns:

- Predicted category
- Detected language
- Confidence score for the predictions
- A short response based on the detected category

2. Traditional Machine Learning

I intentionally used **traditional Machine Learning instead of Deep Learning, Transformers, or LLM-based models**.

The main reasons were:

- The datasets are relatively small and custom-built.
- The problem is primarily a text classification task.
- TF-IDF works effectively for this type of problem.
- Traditional ML models are lightweight and require fewer computational resources.
- They are faster to train and deploy.
- The approach is easier to understand, evaluate, and explain.
- The project demonstrates core NLP and Machine Learning concepts without relying on pre-trained LLMs.

3. Category Classification Model

A custom dataset containing **10,000 samples** was created, with approximately **1,000 samples per category**.

Different algorithms were tested using TF-IDF features:

Algorithm	Accuracy
Logistic Regression	87.45%
Support Vector Machine	~85%
Multinomial Naive Bayes	~99%

Multinomial Naive Bayes achieved the best performance and was selected as the final category classifier. GridSearchCV was used for hyperparameter tuning.

4. Language Classification Model

A separate custom dataset containing **1,200 samples** was created:

- English: 600
- Hinglish: 600

The language classifier uses **TF-IDF + Multinomial Naive Bayes**.

The model achieved:

- **Accuracy: 99.83%**

The model was also tested using additional unseen English and Hinglish sentences and produced correct predictions with high confidence.

5. Validation and Data Leakage Checks

Several checks were performed to verify model reliability:

- Training vs. testing accuracy was compared.
- Train-test accuracy gap was checked.
- 5-fold Stratified Cross-Validation was performed.
- Exact duplicate check: **0 duplicates**.
- Train-test text overlap: **0 samples**.
- Normalized duplicate check: **0 duplicates**.
- Additional unseen sentences were tested manually.

The results showed stable performance with **no significant evidence of overfitting or straightforward data leakage**.

6. Final Technology Approach

User Input

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Language Classifier

TF-IDF + Naive Bayes

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English / Hinglish

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Category Classifier

TF-IDF + Naive Bayes

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One of 10 Categories

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Confidence + Response

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Flask API / Web UI

The trained models are saved using joblib and integrated into the Flask backend for real-time predictions.